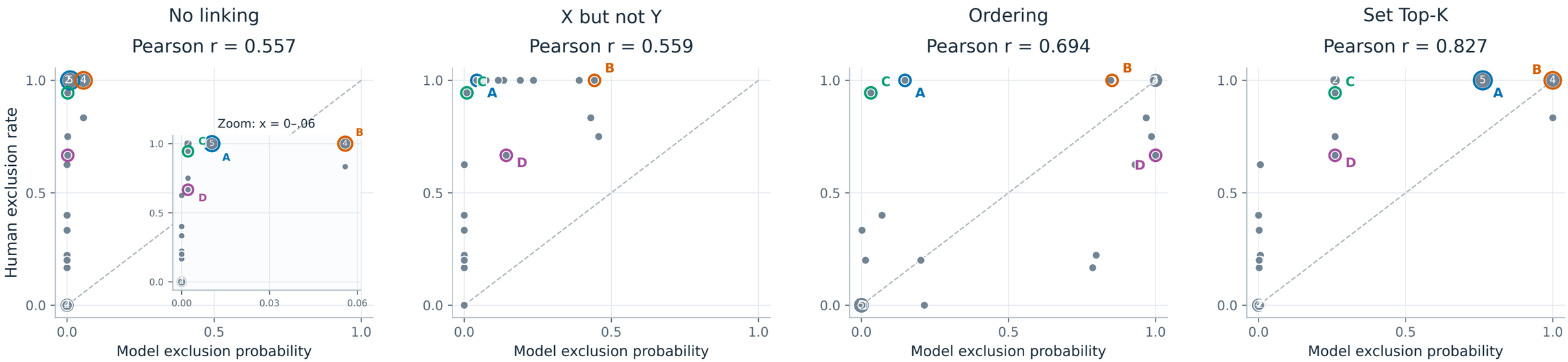
Fridge: linking changes probabilities and sometimes ranks

No linking and Set Top-K have identical rank plots, but different probability spacing and Pearson correlations.

Context: “I’m thirsty.” Sam opens the fridge and responds, “Sure, I have …”

Highlighted pairs

A water → juice B juice → water C water → milk D meat → milk



Fridge fit measures

Fit measure	No linking	X but not Y	Ordering	Set Top-K
Mean log score	-2.971	-1.701	**-0.840**	-1.007
Pearson r	0.557	0.559	0.694	**0.827**
Spearman ρ	**0.860**	0.816	0.769	**0.860**

Each panel includes all 30 pairs with equal pair weights. Numerals inside circles count coincident pairs; colored outlines mark A–D.

Ranks start at 0 (highest exclusion), with average ranks for ties. Dashed lines mark equal values or ranks. Bold table entries are highest.

Corrected Qwen2-7B results · Set K = 20 · natural-log scores with 10^{-10} clipping · descriptive estimates, without uncertainty intervals.